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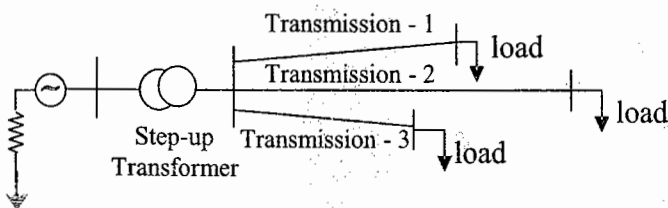
Transmission & Distribution

2.1 Basic Concepts and Line Constants in Transmission

Necessity of Transmission Lines:-

Bulk amount of powers Generation could able to produce on economical basis by employing synchronous Generators at remote locations. The Bulk amount of power from the remote Generating station could able to be carried out to the load center by using suitable network and the network is called transmission network (or) transmission lines.

Transmission line: It is a carrier between the remote Generating station to the load center and it is made by conducting material (e.g. Al, Cu).



Level of voltages:

Low voltage: 220V (1-phase,2-wire),
415V (3-phase,4-wire).

High Voltage : 11kV, 33kV.

Extra High voltage: 66kV, 132kV, 220kV.

Modern Extra High Voltage : 400kV.

Ultra High voltage: 765kV and above.

Most of power generation in India at 11kV.

Maximum AC transmission Voltage in India is 765kV. (e.g. ,Sipat- Seoni 765kV line).

Maximum DC transmission Voltage in India is 500kV.(e.g. Talcher- Kolar Bipolar HVDC link).

Necessity of Extra High Voltages for transmission system:

1. The size of the conductor is reduced so that the cost of the conductor is reduced.
2. The transmission line loss will be reduced.
3. The transmission efficiency will increase
 $\eta = 1 - k / v \cos\phi$

Selection of conductor size for different

Voltage levels:

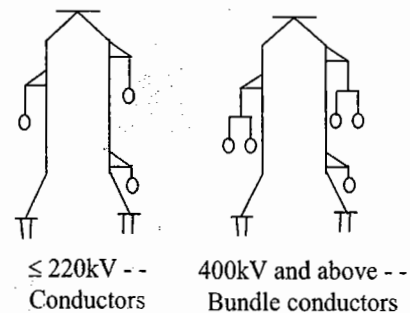
Feeder: It is a conductor having constant current Density. The size of feeder designed based on current carrying capacity. Example for feeder is transmission line on which no tapping are taken to supply any local load.

For $V \leq 220$ kV, selection of conductor done based on current carrying capacity

For $V > 220$ kV, selection of conductor is done based on concept of corona (or) electric field Intensity.

Distributor: It is a conductor having variable current density.

Size of distributor designed based on voltage drop.



Types of conductors:

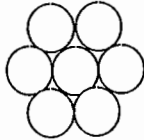
1. Solid Conductor:



Solid

- (i) High Cost
- (ii) High Tensile Strength
- (iii) Difficult to string the conductors.
- (iv) High skin effect while using on ac systems

- 2. Stranded Conductor:** It consists of two or smaller cross sectional strands (or) filaments which are twisted together to get the required strength and running in parallel to increase the current carrying capacity for the given operating voltage.



Stranded

Homogenous Stranded Conductor:-

It is made up of all the strands by one material only i.e., Aluminum.

For AAC (All Aluminum Conductors):

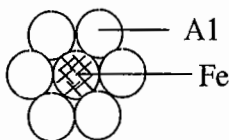
- (i) Reduced Tensile strength due to strands when compared to solid conductor.
- (ii) Easy to string the conductors
- (iii) Reduced skin effect when compared to solid.

But the cost of the conductor is not reduced to greater extent even though the strands are employed.

Due to low tensile strength for conductor, mechanical sag increases and ground clearance of conductor reduces, safety of the system reduces.

In order to improve the tensile strength and reduce the cost of transmission system *composite stranded conductors* are employed (i.e) two (or) more conducting materials used to make stranded conductor.

3. Composite Stranded Conductor:



In the composite stranded conductor the outer strands are having high conductivity and low tensile strength i.e; Aluminium and central strands are having low conductivity and high tensile strength i.e, steel are used which is known as *ACSR conductor*.

ACSR:

1. Steel reinforced Aluminum conductor
2. Aluminum conductor and steel reinforced

The over all Diameter of ACSR conductor will be

$$D = (2n-1) d$$

n = no of layers in the stranded conductor.

d = diameter of each strand in cm.

The number of strands in each layer will be

$$1 + 6 + 12 + 18 + 24 + 30 + 36 \dots$$

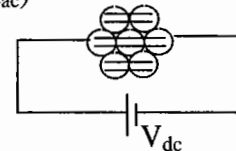
Eg1: An ACSR conductor consists of 30 Aluminum strands and 7 steel strands will be represented as 30/7. In general for a given ACSR conductor the number of Al strands are more in number when compared to steel strands.

Eg2: The ACSR conductor represented as 1/6 It consists of one steel and six Aluminum strands.

Eg3: An ACSR conductor consists of 7 steel strands and 54 Aluminum strands will be represented as 7/54.

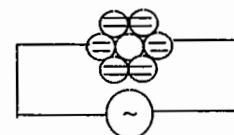
Skin effect:

The non uniform distribution of current through the given cross sectional area of the conductor when it is operated on alternating current system is called skin effect. The main reason for the skin effect is *non-uniform distribution of flux linkages*. The skin effect will result in increased effective resistance (R_{ac})



In case of dc system, there is no rate of change of current. So the current is uniformly distributed throughout the cross sectional area of the conductor. So the entire area will be considered while calculating the **dc resistance** offered.

$$R_{dc} = \frac{\rho l}{a}$$



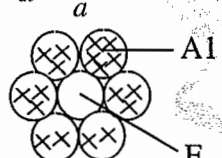
In case of ac system, there is a rate of change of current. Whenever the switch is closed, the system is under sub/transient behavior. So the flux produced in the outer strands will link with the inner strands due to low reluctance offered, where as the flux produced by inner strands could only link with those strands and not with the external strands. **So the internal flux linkages will increases** which will increases the internal inductance.

Inductance will oppose the flow of current, so that the most of the current will concentrated on outer strands and very less current will be allowed through the inner strands. While calculating the effective resistance, the area in which the current concentrated is more only considered than that of entire area of cross section of the conductor. So the ac (or) effective resistance $R_{ac} = \frac{\rho l}{a'}$ ($a' < a$)

R_{ac} (ac resistance) is greater than R_{dc} resistance

$R_{ac} > R_{dc}$ [$\therefore R_{ac} = K R_{dc}$; $K = 1.6$]

In case of ACSR conductor, the current distribution is non-uniform and the materials are also non-uniform. So the entire area will be considered while calculating the effective resistance.

$$R_{ac} = \frac{\rho l}{a}$$


a = Entire area of cross section
(or)

High current is concentrated on the surface. So the conductivity of aluminum is extracted effectively. Less current is concentrated at the inner strands. So the conductivity of steel will also be effectively utilized. Hence the entire cross-sectional area will be considered, while calculating the ac resistance.

- As the diameter (or) Area of cross section of the conductor increases the skin effect will increase i.e., R_{ac} will increase.

- As the permeability of the material will increase, the skin effect will increase i.e., R_{ac} will increase.
- As the operating frequency will increases, the skin effect will increase i.e., R_{ac} will increase. But it is more effective in communication circuits than that of power circuits, because the operating frequency is only 50 Hz.

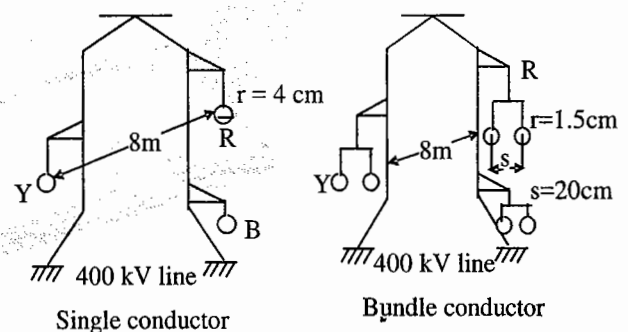
$$\text{Skin effect } (R_{ac}) \propto A \mu_r f$$

$$\text{Skin effect } (R_{ac}) \propto d^2 \mu_r f$$

Bundle conductors:

Whenever the operating voltages beyond 270kv, it is preferable to use more than one sub conductor / phase which is known as *bundle conductor*.

Bundle conductor is one which consists of two (or) more sub conductor / phase which are running in parallel in which the spacing between sub conductors is very high when compared to radius of each sub conductor; where as the distance of separation between the two phases will be very high when compared to the spacing between sub conductors. The each sub conductor is again composite stranded conductor.



Due to sub conductors the mutual distance is same (i/e) 8m. The self distance will increase.

Advantages of Bundle Conductor system:

- Voltage gradient (or) field intensity will

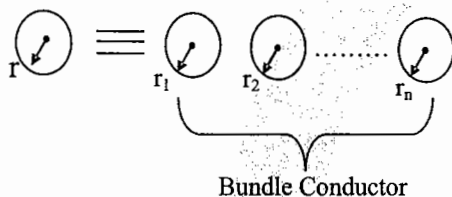
$$\text{reduced } g = E = \frac{V}{\text{GMR} \ln \left(\frac{\text{GMD}}{\text{GMR}} \right)}$$

GMD is same (mutual distance)

GMR is increased (self distance)

- (ii) Due to reduced field Intensity the critical disruptive voltage will increase. So the corona loss is reduced.
- (iii) The reduced corona loss will reduce the communication interference with the adjoining communication lines.
- (iv) Due to increased GMR, Inductance/phase will reduced and capacitance / phase will increase.
- (v) Characteristic impedance will reduced.
- (vi) Characteristic impedance loading will increase.

If the single conductor is converted in to number of sub conductors then the radius of each sub conductor will be calculated as follows.



$$r^2 = r_1^2 + r_2^2 + \dots + r_n^2$$

If all sub conductors are identical

$$r^2 = r_1^2 \cdot n$$

Radius of each sub conductor, $r_1 = \frac{r}{\sqrt{n}}$

for 2 – sub conductors, $r_1 = \frac{r}{\sqrt{2}}$

for 3 – sub conductors, $r_1 = \frac{r}{\sqrt{3}}$

Selection of Insulation level:

- In modern EHV lines the level of Insulation is designed based on switched voltages.
- The internal voltages (switched voltages) will be more severe to the Insulation.
- The insulation will be designed for peak switched voltages.
- Transmission line will be protected from lightning surges by placing a ground wire or shield wire or earth wire on top of the tower.
- Power system network will be protected from switching surges with Surge diverter or Surge Arrester (Non- linear resistor).

Geometric Mean Radius (GMR):

The transmission line conductor having only one conductor per phase, the GMR is the distance between the center to the circumference i.e. it is equal to radius of the conductor. However due to internal flux linkages it is equal to r' .



$$GMR = r' = 0.7788 r$$

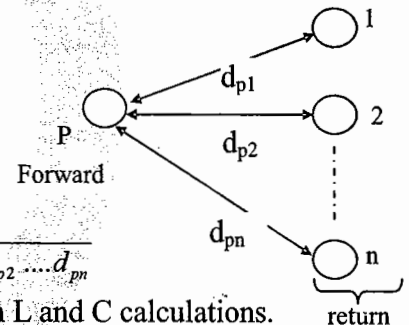
It is also known as self distance.

GMR or self distance used to calculate L of transmission line.

Self GMD is same as GMR and it will be used as a replacement for GMR in multi conductor system.

Geometric Mean Distance (GMD):

If a forward conductor 'p' is surrounding to the 'n' number of return conductors then



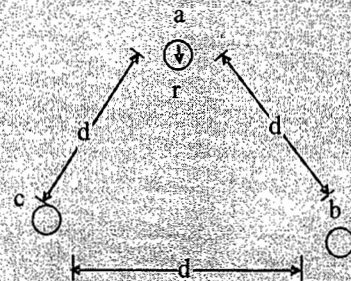
$$GMD_p = \sqrt[n]{d_{p1} \cdot d_{p2} \cdot \dots \cdot d_{pn}}$$

GMD is used in L and C calculations.

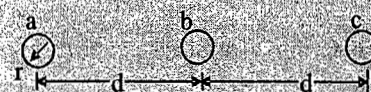
Example 2.1:

Find GMD and GMR of the following systems

(i)



(ii)



$$r = 3 \text{ cm, } d = 4 \text{ m}$$



Sol: (i) $GMR = \sqrt[3]{GMR_a \cdot GMR_b \cdot GMR_c}$
 $= GMR_a$
 $(\because \text{All three conductors are identical})$
 $= r^1$
 $= 0.7788 \times 3 \text{ cm} = 2.34 \text{ cm}$
 $GMD = \sqrt[3]{D_{ab} \cdot D_{bc} \cdot D_{ca}}$
 $= \sqrt[3]{4 \times 4 \times 4} = 4 \text{ m}$

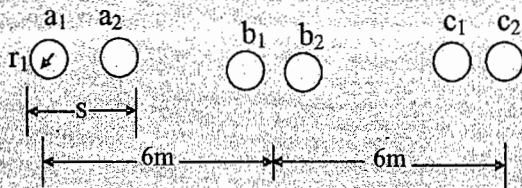
(ii) $GMR = r^1 = 0.7788 \times 3$

$GMD = \sqrt[3]{D_{ab} \cdot D_{bc} \cdot D_{ca}} = \sqrt[3]{4 \times 4 \times 8} = 5.04 \text{ m}$

Observation: GMD of unsymmetrical spacing is more compared to its equivalent symmetrical spacing.

Example 2.2:

Find GMD and Self GMD of the following Bundle conductor system in which $r_1 = 2 \text{ cm}$, $S = 20 \text{ cm}$, all the three phases are identical to each other.



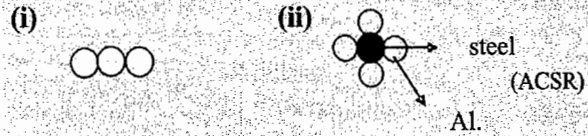
Sol: Self GMD $= \sqrt[3]{\text{selfGMD}_a \cdot \text{selfGMD}_b \cdot \text{selfGMD}_c}$
 $= \text{Self GMD}_a$
 $(\because \text{All three phases are identical})$
 $= \sqrt{r_1 \times S}$
 $= \sqrt{0.7788 \times 2 \times 20} = 5.58 \text{ cm}$

$GMD = \sqrt[3]{6 \times 6 \times 12} = 7.56 \text{ m}$

(By neglecting spacing between sub conductors)

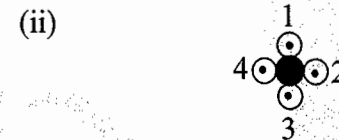
Example 2.3:

Find Self GMD of the following stranded conductors



Each strand radius is $r \text{ cm}$

(i)
Self GMD
 $= \sqrt[3]{(r' \times 2r \times 4r) \times (r' \times 2r \times 2r) \times (r' \times 4r \times 2r)}$
 $= \sqrt[3]{r^9 \times (0.7788)^3 \times 8 \times 4 \times 8}$
 $= r \sqrt[3]{(0.7788)^2 \times 8 \times 4 \times 8}$
 $= 1.75 r \text{ cm}$



The effect of steel strand will be neglected in self GMD calculation.

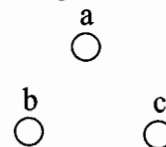
Self GMD of all four strands is same,

Self GMD₁ $= \sqrt[4]{r' \times \sqrt{8}r \times 4r \times \sqrt{8}r}$
 $= \sqrt[4]{r^4 \times 0.7788 \times \sqrt{8} \times 4 \times \sqrt{8}}$
 $= r \sqrt[4]{0.7788 \times \sqrt{8} \times 4 \times \sqrt{8}}$
 $= 2.23 r \text{ cm}$

So, self GMD of conductor also.

Transposition of lines:

It is employed to eliminate the communication interference when transmission line is running close to the communication line. The communication interference is magnetic effect. Normally in transmission lines there are two conductor configurations.



(Symmetrical untransposed)

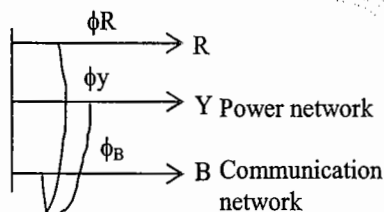


(Unsymmetrical untransposed)

S. No	Line Configuration	Inductance/phase	Voltage /phase	Interference
1	Untransposed symmetrical line	Same	Same	Does not exist because $\phi_r = 0$.
2	Untransposed asymmetrical line	Unequal	Unequal	Exists because $\phi_r \neq 0$.
3	Transposed unsymmetrical line	Equal	Equal	Eliminated because $\phi_r = 0$.

The concept of transposition of lines is considered if the load is balance. However it is a old concept and not suitable for modern power system. The inductance/phase of untransposed unsymmetrical will be the *average inductances of the three phases*.

Definition: It is defined as change in position of power conductor at regular intervals at equal distances throughout the line length, so that the original position of power conductor will be replaced by its successive power conductor.
Untransposed unsymmetrical 3- ϕ transmission line



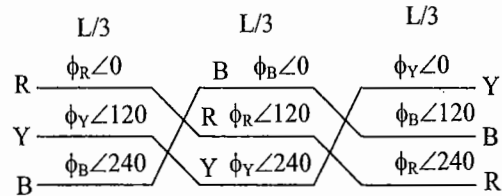
$$\phi_r = \phi_R \angle 0 + \phi_Y \angle 120 + \phi_B \angle 240 \neq 0$$

$$e = \frac{d\phi_r}{dt}$$

$$i_c = \text{circulating current} = \frac{e}{R}$$

R = resistance of communication network

The circulating current will provide the noise in the communication network which is known as communication interference.

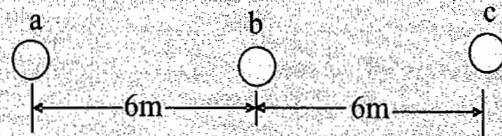


$$\phi_r = \phi_R \angle 0 + \phi_Y \angle 120 + \phi_B \angle 240 + \phi_B \angle 0 + \phi_R \angle 120 + \phi_Y \angle 240 + \phi_Y \angle 0 + \phi_B \angle 120 + \phi_R \angle 240 = 0$$

The transposition of lines means the shifting of the position of power conductor by 120° .

Example 2.1.4:

Find GMD of each conductors and the system for the following conductors configuration without Transposition and with Transposition.



Sol: (i) Without Transposition,

$$GMD_a = \sqrt{6 \times 12} = 8.48 \text{ m}$$

$$GMD_b = \sqrt{6 \times 6} = 6 \text{ m}$$

$$GMD_c = \sqrt{6 \times 12} = 8.48 \text{ m}$$

$$GMD_{sys} = \sqrt[3]{GMD_a \cdot GMD_b \cdot GMD_c} = 7.56 \text{ m}$$

(ii) with transposition,

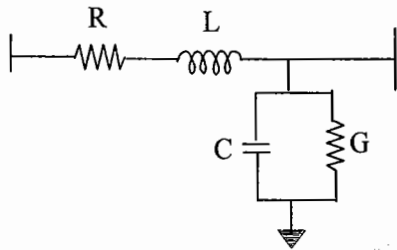
$$GMD_a = GMD_b = GMD_c = \sqrt[3]{6 \times 6 \times 12}$$

$$GMD_{sys} = 7.56 \text{ m}$$



2.2 Transmission Line Parameters

The Transmission line consists of a series combination of resistance, Inductance and a parallel combination of capacitance conductance.



In general the conductance of transmission line is assumed to be zero. The conductance is the one which will be offered for the flow of leakage current through the insulation to the ground. This is only possible at tower location. In between two towers there will not be any leakage.

Resistance Calculation:

It is expressed as

$$R = \frac{\rho l}{a} \Omega/\text{km}.$$

In case of transmission line, the distance is in kilometers. So the parameters will be calculated based on per km length.

ρ = specific resistance (Ω - km)

l = length (km)

a = area of cross section (cm^2).

The resistance is independent on the distance between two wires and it depends on the size of the conductor.

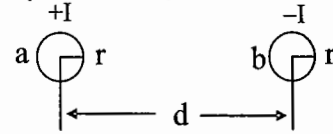
The resistance is also independent on symmetrical / unsymmetrical configuration.

Inductance Calculation:

$$\text{Inductance} = \frac{\text{Magnetic flux linkages}}{\text{Current through conductor}} \text{ H/km}$$

It will produce magnetic field and the energy stored in the inductor $E_L = \frac{1}{2} LI^2$.

Inductance of 1- ϕ , 2- wire system:



The circuit cannot be closed unless there is second wire. The second wire is necessary to maintain potential difference and also closed path for current.

System or loop or total or circuit Inductance

$$L_{\text{total}} = L_a + L_b$$

$$L_{\text{total}} = 2L_a$$

(Both the conductors are similar in size)

Inductance/conductor (L_a) = $L_{\text{int}} + L_{\text{ext}}$

$$L_{\text{int}} = \frac{\mu_0 \mu_r}{8\pi} \text{ and } L_{\text{ext}} = \frac{\mu_0 \mu_r}{2\pi} \ln\left(\frac{d}{r}\right)$$

d = Distance of separation in m

r = radius of conductor in cm

For external Inductance $\mu_r = 1.0$ and for internal Inductance $\mu_r > 1$ except for Cu and Al conductors

$$L_a = \frac{\mu_0 \mu_r}{8\pi} + \frac{\mu}{2\pi} \ln\left(\frac{d}{r}\right) \text{ H/m}$$

$$= 0.05\mu_r + 0.2 \ln\left(\frac{d}{r}\right) \text{ mH/km}$$

In case of Cu or Al conductors $\mu_r = 1.0$

$$L_a = 0.2 \ln\left(\frac{d}{r}\right) \text{ mH/km}$$

r' = imaginary (or) fictitious radius due to addition of internal linkages to external linkages

$$r' = 0.7788r$$

If both conductors are identical then

$$L_{\text{total}} = 2L_a$$

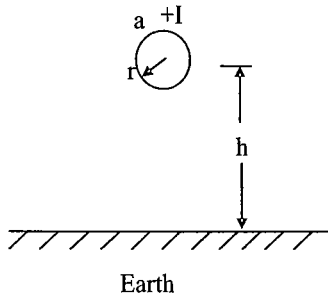
$$L_{\text{total}} = 0.1\mu_r + 0.4 \ln\left(\frac{d}{r}\right) \text{ mH/km}$$



$$L_{\text{total}} = 0.4 \ln\left(\frac{d}{r}\right) \text{ mH/km.}$$

Inductance of 1- ϕ , 1-wire system:

Some times earth will also be considered as return conductor instead of conductor return. But there are certain disadvantages when it is used an a.c system.



Inductance of conductor-a

$$L_a = 0.2 \ln\left(\frac{h}{r}\right) \text{ mH/km}$$

$L_b = 0$, as the radius of earth is very large.

$$\text{So } L_{\text{total}} = L_a + L_b = L_a.$$

Inductance of 3-phase 3-wire Transmission line:

For any 3-phase conductor configuration with transposition, Inductance of each conductor will be same and L/ph is L of any conductor.

$$L/\text{ph} = 0.2 \ln\left(\frac{\text{GMD}}{\text{GMR}}\right) \text{ mH/km.}$$

Inductive reactance per phase, $X_L/\text{ph} = 2\pi f L/\text{ph}$.

Reactive power absorbed by line

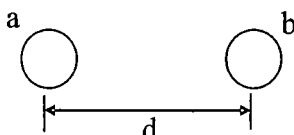
$$Q_L/\text{ph} = I^2 x_L/\text{ph}$$

$$Q_L(3-\text{ph}) = 3I^2 x_L/\text{ph}$$

Example 2.2.1:

A two conductor 1 - ϕ line operate at 50Hz. The diameter of each conductor is 20 mm and the spacing between conductors is 3m. what is the inductive reactance of each conductor in per km.

Sol:



$$\begin{aligned} L_a &= 0.2 \ln\left(\frac{d}{r}\right) \text{ mH/km} \\ &= 0.2 \ln\left(\frac{3}{0.7788 \times 10 \times 10^{-3}}\right) \\ &= 1.19 \text{ mH/km} \end{aligned}$$

$$\text{Inductive reactance } X_a = 2\pi f L_a = 0.374 \text{ } \Omega/\text{km}$$

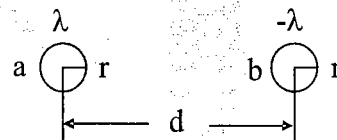
Capacitance Calculation:

$$\text{Capacitance} = \frac{\text{Electric flux linkages}}{\text{Voltage}} \text{ F/km}$$

The importance of capacitor is to develop static field and also to store the energy $E_c = \frac{1}{2} CV^2$.

Capacitance of 1-phase 2-wire system (Isolated earth plane):

The two-wires will act as two plates with a potential difference V or the applied voltage between the two conductors is V .



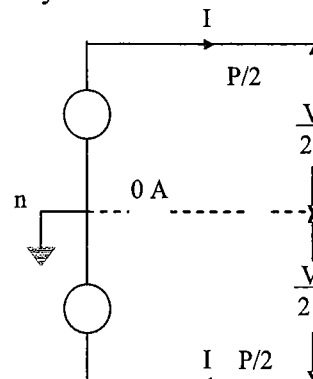
$$C_{ab} = \frac{\lambda}{V} = \frac{\pi \epsilon_0 \epsilon_r}{\ln\left(\frac{d}{r}\right)} \text{ F/m,}$$

$\epsilon_r = 1.0$ for surrounding dielectric medium (air)

$$I_c = V \omega C_{ab} \text{ A/km}$$

Capacitance of each conductor to neutral:

There is a zero potential between the two wires at mid point so that each power conductor will act individually and deliver 50% of the total power.





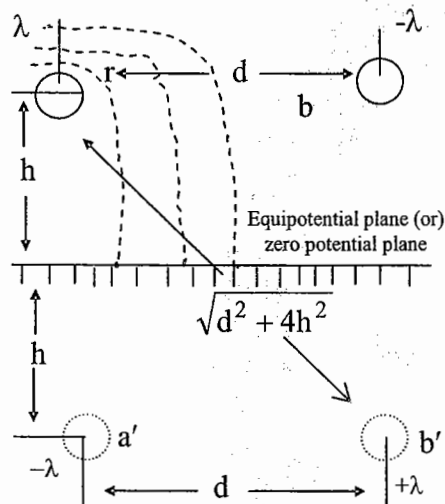
$$C_{an} = \frac{\lambda}{\left(\frac{V}{2}\right)} = \frac{2\lambda}{V} = \frac{2\pi\epsilon_0}{\ln\left(\frac{d}{r}\right)} F/m$$

$$C_{an} = 2 C_{ab}$$

In case of the capacitance calculation, the actual radius of the conductor is considered, whereas in case of inductance calculation imaginary radius is used due to internal magnetic flux linkages.

Capacitance of 1- ϕ 2-wire Transmission Line with the effect of earth:

It can be calculated by using Kelvin's images law.



The electric flux lines which are terminated on infinite plane it will not induce any charge on the infinite plane. However if it is terminated on finite plane, it will induce a charge which is opposite to that of the original charge of the conductor. The no of electric flux lines which are eliminated from the surface of the conductor are equal to the amount of charge which is deposited. Each electric flux line will induce a partial amount of charge on the finite plane. So there is a uniform distribution of charges on the finite plane. It is very difficult to evaluate the capacitance due to uniform distribution of charges. The uniform distribution of charges are replaced by single equivalent charge.

In order to make the earth as zero potential plane the equivalent charge is placed below the earth at a depth equal to the height of the conductor so that

it will be a mirror image to the original power conductor.

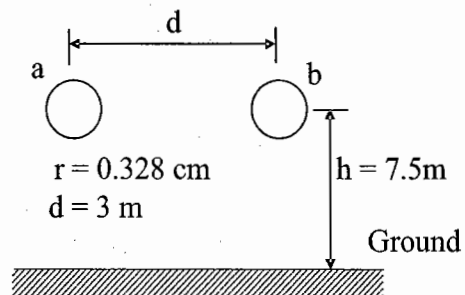
$$C_{an} = \frac{2\pi\epsilon_0}{\ln\left[\frac{d}{r} \frac{2h}{\sqrt{(2h)^2 + d^2}}\right]} F/m$$

Example 2.2.2 :

What is the capacitance of neutral/km of a 1 - ϕ phase line composed of number 2 single strand conductor radius is 0.328 cm, spaced 3 m apart and 7.5 m above the ground? Neglecting earth effect.

- (a) 0.817 $\mu F/km$
- (b) 0.00817 $\mu F/km$
- (c) 0.0212 $\mu F/km$
- (d) 0.516 $\mu F/km$

Sol:



With earth effect

$$C_{an} = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left[\frac{d}{r} \frac{2h}{\sqrt{d^2 + (2h)^2}}\right]} F/m$$

Without earth effect

$$C_{an} = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left[\frac{d}{r}\right]} F/m$$

Earth effect can be neglected,

if $d^2 \ll (2h)^2$ or $d \ll 2h$

$\therefore$ Without earth effect

$$\begin{aligned} C_{an} &= \frac{2\pi \times 8.854 \times 10^{-12} \times 1}{\ln\left[\frac{3}{0.328 \times 10^{-2}}\right]} \\ &= 8.17 \times 10^{-12} F/m \\ &= 0.00817 \mu F/km \end{aligned}$$



Capacitance of 3-phase 3-wire Transmission line:

For any conductor configuration if the conductors are transposed then capacitance of each conductor is same such that.

$$C / \text{ph} = \frac{2\pi\epsilon_0}{\ln \left(\frac{\text{GMD}}{r} \right)} \text{ F/m}$$

Capacitive reactance perphase

$$X_c / \text{ph} = \frac{1}{\omega C / \text{ph}} \Omega - \text{km}$$

Capacitive susceptance, $B_c / \text{ph} = \omega C / \text{ph} \text{ mho-km}$

Charging current, $I_c / \text{ph} = V_{\text{ph}} \omega C / \text{ph} \text{ A/km}$

Reactive power delivered by transmission line

$$Q_c / \text{ph} = \frac{(V_{\text{ph}})^2}{X_c / \text{ph}} = (V_{\text{ph}})^2 \cdot \omega C / \text{ph}$$

$$Q_c (3 - \text{ph}) = 3 \cdot (V_{\text{ph}})^2 \cdot \omega C / \text{ph}$$

Example 2.2.3:

In a 3 phase symmetrical configuration transmission line has inter line capacitance of $0.004 \mu\text{F/km}$. What is the capacitance of each conductor to neutral

Sol: Inter line capacitances in 3 phase configuration connected in Δ to calculate capacitance are each conductor to neutral it should be converted into 'Y'

$$\text{So } C_{\text{an}} = 3 \cdot C_{\text{ab}}$$

$$\text{In this problem } C_{\text{ab}} = 0.004 \mu\text{F/km}$$

$$\text{So, } C_{\text{an}} = 0.012 \mu\text{F/km}$$

Example 2.2.4:

The inductive & capacitive reactance's perphase of a 3 - ϕ , 100km length of line are 50Ω and 1000Ω respectively. What will be the inductive and capacitive reactance of the same line with 50km length.

Sol: Inductive reactance $X_L \propto l$

$$\text{Capacitive reactance } X_C \propto \frac{1}{l}$$

$$\text{For } l = 100\text{km.}, X_L = 50\Omega, X_C = 1000\Omega$$

$$\text{For } l = 50\text{km, } X_L = 25\Omega, X_C = 2000\Omega$$

S.No	Line Description	R	L	X_L	C	X_C
1.	Length increases	Increases	Increases	Increases	Increases	Decreases
2.	Distance of separation increases	No change	Increases	Increases	Decreases	Increases
3.	Radius of conductor increases	Decreases	Decreases	Decreases	Increases	Decreases
4.	Symmetrical spacing	Does not depend	Decreases	Decreases	Increases	Decreases
5.	Unsymmetrical spacing	Does not depend	Increases	Increases	Decreases	Increases
6.	Effect of earth is taken into account	No change	No change	No change	Increases	Decreases
7.	Height of the conductor increases	No change	No change	No change	Decreases	Increases